PHYSICIAN-BUILT · SEER-VALIDATED · N=500

Clinical AI for *early* abdominal cancer detection

An end-to-end pipeline: from presenting symptom to treatment protocol — in under two minutes. Built for primary care settings worldwide where no integrated tool currently exists.

▶ Run Live Demo

GitHub →

MODEL PERFORMANCE · SEER REGISTRY · N=500

ENSEMBLE AUROC (final model)

0.753

BEST SINGLE MODEL — Logistic Regression

0.790

NEGATIVE PREDICTIVE VALUE @ 95% sensitivity

0.853

8.8×

3.7×

STOMACH SURVIVAL GAIN

PANCREATIC SURVIVAL GAIN

WHO IS THIS FOR?

Built for clinicians and researchers

OncosenseAI serves two distinct communities. Select your role to see what matters most to you.

🏥 For Clinicians

🔬 For Researchers

⚡

Triage in under 2 minutes

Enter 17 clinical variables and receive a stratified risk score with a clear referral recommendation — routine, non-urgent, or 2-week-wait — in a primary care consultation.

- → No radiology required to start
- → Alarm symptom weighting built in
- → Outputs match UK/NICE referral thresholds

🌍

Designed for resource-limited settings

Module 2 runs entirely offline with OpenCV — no cloud API, no subscription. Stool, urine, and abdominal image analysis without specialist equipment.

- → Works in LMIC primary care environments
- → No internet dependency for image analysis
- → Optimised for low-bandwidth deployments

💊

Guideline-mapped treatment protocol

Cancer type, stage, ECOG performance, and genomic markers are matched against NCCN, NICE, and ESMO guidelines — including HER2, MSI-H, PD-L1, KRAS, BRAF, and BRCA variants.

- → NCCN Gastric v2.2024 · ESMO 2022
- → Live ClinicalTrials.gov trial matching
- → CheckMate-649 · ToGA · KEYNOTE-590

📋

Shareable clinical PDF report

Module 4 auto-generates a printable clinical summary — patient profile, risk score, imaging findings, treatment recommendation, genomic flags, and matched trials. IRB-ready format.

→ Print-optimised layout

→ Shareable with secondary care teams

→ IRB-ready documentation format

THE PLATFORM

Four integrated modules

From symptom triage to PDF clinical report — a complete end-to-end pipeline. All four modules are complete.

MODULE 01

🧠

Symptom Intelligence Engine

17 clinical variables fed into an LR + GB + RF ensemble. SEER-validated on 500 real cancer patients across 50 years of US registry data.

AUROC 0.790 | 17 VARS | ENSEMBLE | SEER

COMPLETE

MODULE 02

👁️

Visual AI Diagnostics

OpenCV image analysis of stool, urine, and abdominal scans. No external API. Structured JSON output. Offline-capable for LMIC settings.

OPENCV | OFFLINE | JSON OUT

COMPLETE

MODULE 03

Treatment Matcher

Cancer type, stage, ECOG, and genomic markers mapped to NCCN/NICE/ESMO guidelines with live ClinicalTrials.gov API integration.

NCCN/ESMO | HER2 | MSI-H | TRIALS

COMPLETE

MODULE 04

Clinical PDF Report

Auto-generated PDF summarising the full pipeline — risk score, imaging, treatment recommendation, genomic flags, matched trials. Print-ready.

AUTO-PDF | IRB-READY | SHAREABLE

COMPLETE

Publications & preprints

Research outputs from the OncosenseAI project. Preprint submission planned for Q3 2026 following prospective IRB cohort completion.

2026

IN PREPARATION | PREPRINT TARGET: MedRxiv

OncosenseAI: An Integrated Clinical AI Pipeline for Early Detection of Abdominal Cancers in Primary Care Settings

Zuberi R. — Physician-researcher · MBBCh

A four-module end-to-end clinical AI system validated on SEER Cancer Registry data (n=500). Modules cover symptom triage (AUROC 0.790), visual AI diagnostics, guideline-mapped treatment recommendation, and auto-generated PDF clinical reports. Prospective IRB cohort (n=100) in preparation for full validation.

Preprint pending | GitHub →

2026

DATASET · VALIDATED

SEER Registry Validation Study: 12-Month Mortality Prediction in Abdominal Cancers Using Ensemble ML

Zuberi R.

Retrospective validation on 500 SEER Cancer Registry patients. Pancreas (n=264), stomach (n=146), oesophagus (n=90). 5-fold cross-validated ensemble achieves AUROC 0.753 with NPV 0.853 at 95% sensitivity. Survival analysis demonstrates 8.8× median survival advantage for localised vs distant stomach cancer.

Code → | Paper pending

2026 Q2

IRB APPLICATION ACTIVE

Prospective Validation of OncosenseAI in Primary Care: Protocol for a 100-Patient Cohort Study

Zuberi R. · [IRB sponsor — TBD] · [Co-investigators — open]

Prospective cohort validation protocol. 100 patients with suspected abdominal symptoms presenting to primary care. Primary endpoint: sensitivity and specificity of Module 1 risk stratification against gold-standard endoscopic/radiological diagnosis. Co-investigator and IRB sponsorship positions open.

Co-apply →

Co-authorship opportunities are open for the prospective validation study. Contact ramshazubairi81@gmail.com

VALIDATED RESULTS

SEER Registry performance

5-fold cross-validation on real SEER Cancer Registry data. Pancreas (n=264), Stomach (n=146), Oesophagus (n=90). Diagnosis years 2000–2022.

0.790	85.3%
Best AUROC — Logistic	Negative Predictive Value

Regression

Most interpretable model. Ideal for clinical explainability and regulatory audit trails.

At 95% sensitivity threshold — 85% of negatives correctly reassured without unnecessary referral.

500

Validated Patient Records

Real SEER registry data spanning 50 years of US cancer outcomes. Not synthetic data.

MODEL	AUROC	ROLE	CONFIG
Logistic Regression	0.790	Best single model	Interpretable, clinically explainable
Random Forest	0.752	Strong performer	200 trees, max depth 6
Gradient Boosting	0.694	Good	200 estimators, depth 4
✅ Ensemble (deployed)	0.753	Final model	Soft voting, weighted average

Stomach Cancer n=146

Localised — early detection	35 months
Regional	17 months

Distant — how most present today	4 months

8.8×

survival advantage with early detection

Pancreatic Cancer n=264

Localised — early detection	11 months
Regional	12 months
Distant — how most present today	3 months

3.7×

survival advantage with early detection

MODULE 1 · LIVE DEMO

Symptom triage engine

Enter patient details to receive a risk stratification and referral recommendation. Research demonstration only — not for clinical use.

Patient Assessment

// RESEARCH PROTOTYPE — NOT FOR CLINICAL USE

AGE (years)

50

SEX

Select

BMI

24.5

SYMPTOM DURATION

Select

ALARM SYMPTOMS · ● = high specificity

- [ ] Rectal bleeding
- [ ] Unexplained weight loss
- [ ] Jaundice
- [ ] Palpable mass
- [ ] Dysphagia
- [ ] Change in bowel habit
- [ ] Iron deficiency
- [ ] Abdominal pain
- [ ] Early satiety
- [ ] Fatigue

FAMILY HISTORY

No

PRIOR GI DIAGNOSIS

None

Run Risk Assessment →

Complete the form and click
Run to see risk stratification

DEVELOPMENT ROADMAP

From prototype to clinical validation

A clear path from SEER validation through IRB pilot to regulatory submission.

2026 Q1 · COMPLETE

Modules 1–4 built and validated

Full pipeline operational. SEER validation n=500. AUROC 0.790. All modules open-source on GitHub.

2026 Q2 · IN PROGRESS

IRB pilot — 100-patient prospective cohort

IRB sponsorship outreach active. MIMIC-IV application in progress. Co-investigator positions open.

2026 Q3

MedRxiv preprint submission

Peer-reviewed preprint with full methodology, SEER validation, and prospective cohort results. Co-authorship open.

2026 Q4

FDA 510(k) pre-submission meeting

Pre-submission scoping meeting. Regulatory pathway for clinical AI decision support tool.

Seeking collaboration

Built by a physician. Looking for research partners, IRB sponsors, and grant co-applicants to bring this to patients globally.

🏥 US Academic Medical Centre Position

📋 IRB Sponsorship

💰 NIH SBIR / NCI R21

🔬 Oncology · Radiology · Clinical AI · Global Health

📧 **ramshazubairi81@gmail.com**

LinkedIn →

GitHub →

RESEARCH PROTOTYPE ONLY. Not validated for clinical use. Not approved by FDA, CE, or any regulatory authority. Do not use for actual patient care decisions.